Python Data Analysis Assignment 1 Documentation

This document acts as a `README` manual for the completed NumPy and Pandas analytical pipeline scripts.

Technical Solution Summary

1. NumPy Array Manipulations

- **1D Array Inspection:** Created array structures from daily temperatures, calculating shape, dimensions, and standard types.
- **Vector Math Operations:** Utilized high-performance vectorized operations to process absolute Celsius-to-Fahrenheit conversions.
- **Descriptive Metrics:** Extracted core statistical bounds including minimum, maximum, and average values.
- **Interval Slicing:** Sub-sectioned lists to partition early-week, weekend, and mid-week intervals across multi-dimensional sets.

2. Pandas Index Series

- **Label Constraints:** Assigned custom ranking strings mapped directly to numeric records.
- **Accessor Lookups:** Isolated structural queries cleanly separating implicit positions (`.iloc`) from row keys (`.loc`).
- **Logical Bitwise Filtering:** Filtered and matched records meeting specified performance criteria (`marks > 90`).

3. Pandas Tabular DataFrames

- **Data Quality Audits:** Evaluated structures using fast exploration features (`head()` , `tail()` , `shape` , `dtypes`).

- **Cross-Sectional Subsetting:** Queried multi-column sets across mixed logical constraints.
- **Aggregation Splits:** Segmented structural transaction values using grouped operational routines (`groupby()`).

Execution Environment Guide

Ensure you have the required analytical pack tools active inside your working workspace path:

```
pip install numpy pandas
```

To run the execution routine:

```
python assignment.py
```